

Лабораторная работа №11

Управление загрузкой системы (GRUB2)

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Цель работы

Получить навыки работы с загрузчиком системы **GRUB2**, его конфигурацией и методами восстановления загрузки системы.

Ход выполнения работы

Модификация параметров GRUB2



```
rkogengar@rishardkogengar:/home/rkogengar - nano /etc/default/grub
GNU nano 8.1 /etc/default/grub Modified
GRUB_TIMEOUT=20
GRUB_DISTRIBUTOR="$(sed 's, release .*$,,g' /etc/system-release)"
GRUB_DEFAULT=saved
GRUB_DISABLE_SUBMENU=true
GRUB_TERMINAL_OUTPUT="console"
GRUB_CMDLINE_LINUX="resume=UUID=fa00e072-101d-4203-881d-9acf2f98b668 rd.lvm.lv=rl_vbox/root rd.lvm.lv=rl_vbox/swap"
GRUB_DISABLE_RECOVERY="true"
GRUB_ENABLE_BLSCFG=true
```

Рис. 1: Редактирование файла /etc/default/grub

```
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# nano /etc/default/grub
root@rishardkogengar:/home/rkogengar# grub2-mkconfig > /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
root@rishardkogengar:/home/rkogengar# grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
root@rishardkogengar:/home/rkogengar# █
```

Рис. 2: Создание нового файла конфигурации GRUB2

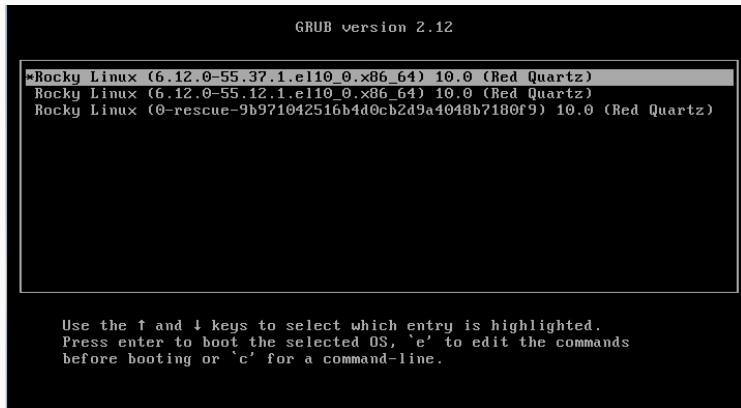


Рис. 3: Отображение меню GRUB

Устранение неполадок

```
GRUB version 2.12

load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-6.12.0-55.37.1.el10_0.x86_64 root=/dev/mapper/rl_vbox\
-root ro resume=UUID=fa00e072-101d-4203-881d-9acf2f98b668 rd.lvm.lv=rl_vbox\
/root rd.lvm.lv=rl_vbox/swap systemd.unit=rescue.target
initrd ($root)/initramfs-6.12.0-55.37.1.el10_0.x86_64.img $tuned_initrd

Minimum Emacs-like screen editing is supported. TAB lists
completions. Press Ctrl-x or F10 to boot, Ctrl-c or F2 for
a command-line or ESC to discard edits and return to the GRUB menu.
```

Рис. 4: Редактирование параметров ядра для rescue.target

Проверка системных модулей

```
systemd-udev-control.socket loaded active 1
systemd-udev-kernel.socket loaded active 1
dev-disk-by\x2duuid-fa00e072\x2d101d\x2d4203\x2d881d\x2d9acf2f98b668.swap loaded active a
cryptsetup.target loaded active a
integritysetup.target loaded active a
local-fs-pre.target loaded active a
local-fs.target loaded active a
network-pre.target loaded active a
rescue.target loaded active a
sound.target loaded active a
swap.target loaded active a
sysinit.target loaded active a
veritysetup.target loaded active a

Legend: LOAD    + Reflects whether the unit definition was properly loaded.
              ACTIVE + The high-level unit activation state, i.e. generalization of SUB.
              SUB    + The low-level unit activation state, values depend on unit type.

69 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
root@rishardkogengar:~# systemctl show-environment
LANG=en_US.UTF-8
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
XDG_DATA_DIRS=/var/lib/flatpak/exports/share:/usr/local/share/:/usr/share/
root@rishardkogengar:~#
```

Рис. 5: Проверка загруженных модулей и переменных окружения

```
GRUB version 2.12

load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-6.12.0-55.37.1.el10_0.x86_64 root=/dev/mapper/rl_vbox\
-root ro resume=UUID=fa00e072-101d-4203-881d-9acf2f98b668 rd.lvm.lv=rl_vbox\
/root rd.lvm.lv=rl_vbox/swap systemd.unit=emergency.target
initrd ($root)/initramfs-6.12.0-55.37.1.el10_0.x86_64.img $tuned_initrd

Minimum Emacs-like screen editing is supported. TAB lists
completions. Press Ctrl-x or F10 to boot, Ctrl-c or F2 for
a command-line or ESC to discard edits and return to the GRUB menu.
```

Рис. 6: Настройка загрузки в emergency.target

```
• sys-devices-pci0000:00-0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda.device
• sys-devices-platform-serial18250-serial18250:0-serial18250:0.0-tty-ttyS0.device
• sys-devices-platform-serial18250-serial18250:0-serial18250:0.1-tty-ttyS1.device
• sys-devices-platform-serial18250-serial18250:0-serial18250:0.2-tty-ttyS2.device
• sys-devices-platform-serial18250-serial18250:0-serial18250:0.3-tty-ttyS3.device
  sys-devices-virtual-block-dmxx2d0.device
  sys-devices-virtual-block-dmxx2d1.device
• sys-module-configfs.device
• sys-module-fuse.device
  -.mount
  sys-kernel-config.mount
  init.scope
  emergency.service
  plymouth-start.service
  systemd-journald.service
  -.slice
  system-modprobe.slice
  system.slice
  systemd-journald-dev-log.socket
  systemd-journald.socket
  emergency.target
```

Legend: LOAD + Reflects whether the unit definition was properly loaded.
 ACTIVE + The high-level unit activation state, i.e. generalization of SUB.
 SUB + The low-level unit activation state, values depend on unit type.

68 loaded units listed. Pass --all to see loaded but inactive units, too.

To show all installed unit files use 'systemctl list-unit-files'.

root@rishardkogengar:~#

Рис. 7: Проверка активных юнитов в emergency.target

Сброс пароля root

```
GRUB version 2.12

load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-6.12.0-55.37.1.el10_0.x86_64 root=/dev/mapper/rl_vbox\
-root ro resume=UUID=fa00e072-101d-4203-881d-9acf2f98b668 rd.lvm.lv=rl_vbox\
/root rd.lvm.lv=rl_vbox/swap rd.break
initrd ($root)/initramfs-6.12.0-55.37.1.el10_0.x86_64.img $tuned_initrd

Minimum Emacs-like screen editing is supported. TAB lists
completions. Press Ctrl-x or F10 to boot, Ctrl-c or F2 for
a command-line or ESC to discard edits and return to the GRUB menu.
```

Рис. 8: Загрузка с параметром rd.break

```
Generating "/run/initramfs/rdsosreport.txt"

Entering emergency mode. Exit the shell to continue.
Type "journalctl" to view system logs.
You might want to save "/run/initramfs/rdsosreport.txt" to a USB stick or /boot
after mounting them and attach it to a bug report.

Give root password for maintenance
(or press Control-D to continue):
sh-5.2# mount -o remount,rw /sysroot/
sh-5.2# chroot /sysroot/
sh: chroot: command not found
sh-5.2# passwd
sh: passwd: command not found
sh-5.2#
```

Рис. 9: Попытка сброса пароля root

Итоги работы

В ходе лабораторной работы были изучены принципы настройки загрузчика **GRUB2**, методы устранения ошибок загрузки и процедуры восстановления системы, включая сброс пароля пользователя root.